Practice 6: Echo State Networks

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Objective

The practical implements a Reservoir Computing (RC)/ Echo State Network (ESN) model applied to neural recordings from a non-human primate. The model is used to classify movement-related brain states from multi-electrode array data (128 channels, sampled at 2048 Hz).

Two classifications are performed:

- **Five State Classification (5s):** distinguish five movement-related states simultaneously, using a full trial as input.
- **Left vs Right (lvr):** distinguish left-arm from right-arm activity, with trials intercalated in a paired order (states 1-5 left paired with states 6-10 right) to avoid asymmetric learning.

Pipeline

1) **Data loading and splitting (`data.py`):** MATLAB data is loaded and split 80/20 into training and test sets. A frequency-band filter is applied before passing the data into the network.
2) **Network setup (`network.py`):** A sparse random reservoir (N nodes) is constructed with a random input matrix (W_in) and a recurrent adjacency matrix (W) whose spectral radius is scaled to below 1 (eigenvalue normalization). This avoids internal instability.
3) **Training (`reservoir.py`):** Input signals are fed through the reservoir. At each trial, the node activations are averaged over the trial window to produce a feature vector per trial. These vectors are then used to fit a readout (either Linear Regression, Logistic Regression, 1-Nearest Neighbor… depending on the task).
4) **Testing:** The same averaging steps are applied to the test set and the fitted readout is evaluated. For instance, for the linear readout, accuracy is computed by taking the argmax over output nodes.

Comparison with Linear AR Models

Linear autorregressive models (AR) capture temporal dependencies in a parametric, rigid manner: they assume stationary and model the signal as a fixed linear combination of past values. Their temporal horizon is determined by the model order.

Reservoir computing provides a complementary approach with several differences:

- **Non-linear dynamics**: the tanh non-linearity in the state update equation allows the reservoir represent non-linear combinations of past inputs, which linear AR models cannot do.

- **Variable, implicit memory:** the reservoir state encodes a fading memory of the input, whose timescale is not fixed a priory. But is the result of the spectral radius and connectivity of W. This is analogous to an adaptive lag selection.
- **Data efficiency:** the only trained components is the linear readout; the reservoir itself is fixed at random. This makes RC suited for datasets with scarce labeled data, avoiding overfitting problems that come when training a large parametric model.
- **High dimensional projection:** even with a simple classifier as output, the reservoir projects the input into a high-dimensional non-linear space. It separates the classes that may be li9nearly inseparable with just raw inputs.

The principal trade-off is interpretability: unlike AR coefficients, reservoir states are not directly interpretable. The choice of spectral radius, sparsity, and frequency band also introduces hyperparameter sensitivity absent in standard AR fitting.

In the neural decoding context of this practical, RC is expected to outperform linear models when the discriminative features involve non-linear or multi-scale structure. This are conditions inherent in movement-related neural activity.